

PARTS LIST

Semiconductors:

- IC1, IC2 - LM1875 audio amplifier
- T1-T3 - BC547C npn transistor
- LED1 - 5mm LED
- D1 - 1N4148 signal diode
- ZD1 - 20V, 1.3W Zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1 - 220-ohm
- R2, R3 - 330-kilo-ohm
- R4, R24, R25 - 10-kilo-ohm
- R5 - 810-kilo-ohm
- R6 - 22-ohm
- R7 - 2.2-mega-ohm
- R8, R9, R15-R17, R19 - 22-kilo-ohm
- R10 - 100-kilo-ohm
- R11 - 120-kilo-ohm
- R12, R13 - 14-kilo-ohm
- R14 - 430-ohm
- R18, R20 - 680-ohm
- R21, R22 - 2.2-ohm, 3W
- R23 - 20-kilo-ohm
- VR1, VR2 - 4.7-kilo-ohm potmeter

Capacitors:

- C1, C4, C11, C13 - 0.03 μ F ceramic
- C2 - 4.7 μ F, 100V non-polar capacitor
- C3 - 2.2nF ceramic
- C5, C9 - 150pF ceramic
- C6, C7, C10, C12 - 4.7 μ F, 63V electrolytic
- C8 - 470 μ F, 63V electrolytic
- C14, C15 - 1000 μ F, 63V electrolytic
- C16, C17 - 22 μ F, 63V electrolytic
- C18, C19 - 0.22 μ F ceramic
- C20, C21 - 2200 μ F, 63V electrolytic
- C22 - 100 μ F, 63V electrolytic

Miscellaneous:

- S1 - Single-pole, two-way (SPDT) switch
- CON1-CON7 - 2-pin connector terminal
- LS1-LS2 - 4-ohm, 20W loudspeaker
- LS3 - 8-ohm, 40W loudspeaker
- 30V, 3A DC power supply
- Heat-sink for IC1 and IC2
- Microphone
- Guitar

above 100 and depends on potmeter VR1 and resistor R6.

CON2 and CON3 are for inputs to guitars. Resistance of a single input is above 1-mega-ohm.

Values of R2 and R3 are selected according to the guitar pickups. T2 works as emitter follower with gain close to unity. T1, T2 and T3 should be of low-noise and high-gain type such as BC109C, BC547C or BC550C.

Signals from all inputs are summed in T3 stage, which provides two output signals with the same amplitude but with opposite phases. These signals are applied

TABLE II
TEST POINTS

Test point	Details
TP0	0V (GND)
TP1	30V
TP2	20V
TP3	12V
TP4-TP6	15V

to two identical output stages built around IC1 and IC2.

Switch S1 gives the option to select the same or opposite signals for the stage using IC2. Both output stages are identical. These can be driven with the same or with opposite signals. Gain of the output stage is around 33 and can be readjusted with feedback resistors as needed.

Mode-control switch S1. The amplifier has several configurations depending on the position of S1 and connection of loads. S1 has two positions, 1 and 2.

When S1 is in position 1, both output stages have the same input signals. Loads have two configurations. In the first configuration, two separate loads (loudspeakers) can be connected to CON5 and CON6.

In the second case, CON5 and CON6 can be paralleled through small equalisation resistors (usually 0.05-0.1-ohm load resistance) and a common load can be applied.

This is possible because output stages have very close parameters. Gain-control resistors should be matched to better than $\pm 1\%$. Care must be taken about polarity of signals and polarity of connections of loudspeakers.

When S1 is in position 2, output channels are driven by opposite signals. Here also, there are two possible configurations for loads. In the first case, two separate loads can be connected to CON5 and CON6.

In the second case, CON5 and CON6 are not used, and only a single load is connected to CON7.

In both cases, load can be from 4-ohm to 16-ohm (sometimes even

more), and its resistance and power depends on the ICs selected from Table I.

Power supply. DC power supply is applied to CON4 as per power amplifier ICs selected from Table I. Power supply can be regulated or unregulated.

Values of capacitors C11, C13, C14 and C15 can be increased or paralleled with additional capacitors. Power supply for the transistors is additionally filtered and regulated with Zener diode ZD1, capacitors C1 and C8 and resistor R14. Zener ZD1 is BZX85 type with maximum power of 1.3W, depending on the selected power supply.

Heat-sink. IC1 and IC2 are mounted on appropriate common heat-sinks, depending on the required output power. It is preferable to use heat-sinks with thermal resistance lower than 1.5°C/W. Bigger heat-sinks are better.

Construction and testing

An actual-size, single-side PCB pattern for the simple dual-way amplifier is shown in Fig. 2 and its component layout in Fig. 3.

The loudspeakers can be 4-ohm to 16-ohm, peak-output current can go up to 3A or even 5A in some cases, and peak-output power is from 10W to more than 40W, depending on the selected ICs, heat-sinks, power supply and configuration of output (dual-way or bridged amplifier).

This power amplifier usually does not need any adjustment, but it is important to have the gains of the output stages as mentioned earlier. For troubleshooting, check voltages at various test points listed in Table II. **EFY**



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